

Lemma Academy Educational Consultancy

Project Kaieteur

Proposal to Guyana Ministry of Education



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Project Kaieteur

National Robotics and Technological Education Strategy Proposal

Synopsis

A systematic approach to the development of STEM and robotics education in Guyana is proposed. Building on existing STEM programs within the Ministry of Education, small cadres of willing and enthusiastic elementary and secondary teachers are to be trained, offered technological support and mentorship and empowered to start STEM activities including *FIRST* Robotics Competition teams. Investment in the program by the Ministry of Education is to be enhanced and leveraged through the engagement of the Guyanese diaspora in the United States and Canada as well as the private sector in Guyana.

Objective of the Proposed Project

The Motto and Vision Statement of the Ministry of Education of Guyana are as follows:

Motto: *Eliminating illiteracy, modernizing education, strengthening tolerance*

Vision Statement: *To be a Ministry of Excellence in providing quality education and promoting arts and culture for the prosperity of the Nation*

The **objective** of the proposed project is intended to harmonize with the above motto and vision statement:

To foster a culture of innovation, collaboration, and STEM excellence across Guyana by establishing affordable, sustainable and engaging STEM activities in every school—creating in the process dynamic communities of highly motivated and capable youth prepared to support the ongoing rapid transformation of Guyana's economy.

The Proposal in Brief

The key lines of action of this proposal appear in the chart below. Where possible, the lines of action have been proposed in such a way that they may build upon efforts already made by the Ministry of Education of Guyana and infrastructure and technologies that already exist within the Ministry and the nation's schools.

The core of the proposal is encapsulated by **Lines of Action 1 and 2** which are to be carried out directly by Lemma Academy Education Consultancy with the support of the Ministry of Education of Guyana. The remaining five Lines of Action are proposed to ensure that Lines of Action 1 and 2 are effective and sustainable and that their impact is measurable. These



will involve collaboration with other individuals and organizations including the Ministry of Education of Guyana, post secondary institutions and individuals of capacity representing various industries in Guyana and from the Guyanese diaspora.

#	Line of Action	Description
1	Training Workshops	• Hold Training Workshops for Principals, Teachers and Students • Offer elementary and secondary principals and teachers throughout the nation an opportunity to apply for a professional development opportunity in STEM and Robotics • Workshop 1 - August, 2025 - Select small groups of principals and teachers from 2-3 <i>elementary</i> schools and 2-3 secondary schools from each of Guyana's ten regions • Workshop 2 - December, 2025 - Select small groups of teachers and students from 2-3 elementary and 2-3 <i>secondary</i> schools from each of Guyana's ten regions • Subsequent to the workshops, support may be periodically be offered online
2	STEM Activities	• Start elementary and secondary school STEM-related activities with due regard to the available finance, human resources and capabilities in each school and region. ◦ Block Coding ◦ Python Computer Programming (Free resource) ◦ MakeBlock Robots ◦ Blender 3D design (Free resource) ◦ CAD (FreeCAD / Onshape - <i>Free</i> resources) ◦ 3D Printing \$ ◦ <i>FIRST</i> Lego League Challenge classpacks and teams (ages 9-16) \$ ◦ Arduino (\$) ◦ Raspberry Pi (\$\$) ◦ <i>FIRST</i> Tech Challenge teams(ages 12-18) \$\$ ◦ Start a National <i>FIRST</i> Robotics Competition Team to represent the country internationally, offering spots to the best and brightest youth in the country \$\$\$
3	Communication	• Start a YouTube Playlist specific to Project Kaieteur ◦ Share clips from classrooms and from STEM and Robotics events throughout the nation • Write a newsletter to the teachers of the nation every 1-3 months encouraging teachers and students with

		news of various developments related to the project ○ Encourage principals and teachers with the knowledge that they have the full support of the Ministry of Education in their individual initiatives ○ Share best practices from throughout the country ○ Celebrate innovations, events and various other positive developments ● Advertise the project on radio or television to create a sense new developments are taking place nationwide
4	Add subjects and events to the Guyana STEAM Fair	● Build upon the existing Guyana STEAM Fair with coding and robotics activities ● This platform provides an exciting motivator tying together efforts from throughout the country and bringing the curriculum to life! ● Suggested additional event categories correspond with the STEM activities initiated through Line of Action 2: ○ Coding ○ Robotics ■ Engineering ■ CAD ○ Mathematics
5	Leverage Available Networks, Infrastructure and Resources	● Support the Creation of a national not-for-profit robotics education agency to support the establishment of <i>FIRST</i> Robotics teams throughout the country ● Reach out to key ministries in the Guyana Government: ○ Ministry of Amerindian Affairs - collaboration in reaching out to communities in the interior ○ Guyana Revenue Authority - to request tax exemption for import of parts and equipment related to robotics education ● Call on key individuals in the private sector and Guyanese diaspora to support the project financially and through their expertise ● Engage post secondary partners to garner support for the program and expertise ○ University of Guyana ○ Guyana Technical Training College ○ Cyril Potter College of Education ○ Nations University
6	Curriculum	● Revise the elementary and secondary school curricular to reflect the emergence of STEM and technological education

		• Integrate lessons into math, science and social studies classes at the elementary level • Introduce new robotics and technological education courses at the secondary school level
7	Metrics	• Devise and Implement Metrics to Gauge the Impact of the Program, building on existing metrics used by the Ministry of Education • Begin with the mission and vision of the Ministry of Education: ◦ <i>To be a Ministry of Excellence in providing quality education and promoting arts and culture for the prosperity of the Nation</i> ◦ <i>Eliminating illiteracy, modernizing education, strengthening tolerance</i> • Consider the overall goals and aims of the Ministry of Education with respect to student performance in academics and acquisition of hard and soft skills related to robotics, STEM education and technological education • Collect data through: ◦ Surveys ◦ Analysis of projects created by various classes ◦ Analysis of videos submitted by participating classes ◦ Case studies ◦ Interviews of principals, teachers and students

Implementation Framework

Lasting impact in an education project depends upon achieving a shift in culture. While technological resources such as computers, software packages and robotics kits may be a factor in the success of a project, they are not sufficient to achieve sustainable growth. Long term results are seen when there is a positive shift at the level of culture. Teachers and principals require adequate clarity and structure to have the confidence to undertake new initiatives. Students are motivated by a sense of meaning and encouragement. The content of the training in the initial stages will naturally involve a considerable amount of time spent on creating the right environment for regional education councils, principals, teachers and students to take action within the Framework of the program.